

Ondřej Kubů

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PROFILE

Research mathematician (Severo Ochoa postdoc, ICMAT-CSIC Madrid) moving into technical AI safety. Doctoral training in mathematical physics: differential geometry and integrable systems, with substantial graduate work in operator and spectral theory. 8 peer-reviewed articles, one further article in press, and 1 preprint under review.

AI SAFETY

AGI Strategy course, BlueDot Impact (Jul 2026), certificate. Final project on compute-governance verification through hardware-enabled mechanisms.

Independent study of formal impossibility results in AI control (Alfonseca et al. on containment, Yampolskiy on controllability): checking which proofs hold and which applications of Rice's theorem go through.

Evaluation of theory-first safety agendas, ongoing: worked through ARC's mechanistic-safety programme (incl. Winer's account, Matolcsi's obstacles, and Wu et al. 2026 on cumulant-propagation estimators) and Bengio's Scientist AI when choosing a research direction.

Volunteer, PauseAI CZ, z. s. (2026–present). Author of the Czech-language weekly newsletter *Pozastavení nad AI*, synthesizing frontier-AI developments and their EU and Czech policy implications for a non-specialist audience.

Giving What We Can 10% Pledge – committed to donating 10% of lifetime income to effective charities.

EMPLOYMENT

Dec 2024 – Nov 2026	Postdoc (Severo Ochoa) , Instituto de Ciencias Matemáticas (ICMAT), CSIC, Madrid. Haantjes algebras for separation of variables of (super)integrable systems. Supervisor: P. Tempesta
Sep 2024 – Nov 2024	Research Associate , FNSPE, Czech Technical University in Prague
2021 – 2024	Assistant lecturer , Dept. of Physics, FNSPE CTU. Seminars in Theoretical Physics 2, Lie Algebras and Groups, and Geometrical Methods in Physics.

EDUCATION

2020 – 2024	Ph. D. , FNSPE CTU Prague, Mathematical Engineering / Mathematical Physics. Thesis: <i>Integrability and superintegrability in the presence of magnetic fields: separable and nonseparable systems</i> . Supervisor: L. Šnobl; Co-supervisor: A. Marchesiello. <i>Rector's Award for Excellent Doctoral Thesis (2025)</i> . Grade: 1.00 GPA (Excellent); oral defence Grade 1.
2018 – 2020	Master's Degree with Honors (Ing.) , FNSPE CTU, Mathematical Physics. Supervisor: L. Šnobl; Consultant: P. Winternitz. Grade: A (GPA 1.31).
2015 – 2018	Bachelor's Degree with Honors , FNSPE CTU. Grade: A (GPA 1.41).

SELECTED GRADUATE COURSEWORK

Doctoral courses, all graded 1 (excellent): Schrödinger Operators; Geometrical Aspects of Spectral Theory; Introduction to the Theory of (Operator) Semigroups; Classification and Identification of Lie Algebras; Solvable Models of Mathematical Physics. Earlier coursework in partial differential equations via functional analysis, and in statistical physics.

RESEARCH VISITS

ERASMUS+ research exchanges during studies: Universidad Complutense de Madrid, Spring 2023 (P. Tempesta); Université de Montréal, Fall 2019 (P. Winternitz).

AWARDS	Rector's Award for Excellent Doctoral Thesis , CTU (2025); Josef Hlávka Award for the best students and graduates of public universities in Prague (2023); Milan Odehnal Award (2024), honorable mention, Czech Physical Society; Dean's Award for the 2nd best Master Thesis (2021).
SKILLS	Software: Maple, $\text{\LaTeX}$; author of an open Maple package for Nijenhuis/Haantjes torsion computation, github.com/OndrejKubu/NijenhuisHaantjesTorsion (2026). Basics of C++; daily use of AI coding assistants in research. Service: 4 peer reviews for <i>Journal of Physics A</i>. Languages: Czech (native), English (C1–C2), German (B2/C1), French (B2), Spanish (A2).
PUBLICATIONS	Full list: ORCID 0000-0001-7463-5680. <i>Preprint</i> O. Kubů, D. Latini. Haantjes algebras, Zernike system and separation of variables, under review. arXiv:2605.23748 (2026) <i>Journal articles</i> O. Kubů, P. Tempesta. Stäckel and Eisenhart geometries, symplectic-Haantjes manifolds and gravitation, accepted, <i>Proc. R. Soc. A</i>; in press. DOI: 10.1098/rspa.2025.1103. arXiv:2509.19950 O. Kubů, D. Reyes, P. Tempesta, G. Tondo. Hamiltonian integrable systems in a magnetic field and symplectic-Haantjes geometry. <i>Proc. R. Soc. A</i> 480 (2024) 20240076 O. Kubů, A. Marchesiello, L. Šnobl. Integrable systems with velocity-dependent potentials: generalized parabolic cylindrical case. <i>J. Phys. A</i> 57 (2024) 235203 O. Kubů, A. Marchesiello, L. Šnobl. New classes of quadratically integrable systems in magnetic fields: the generalized cylindrical and spherical cases. <i>Ann. Phys.</i> 451 (2023) 169264 O. Kubů, L. Šnobl. Cylindrical first-order superintegrability with complex magnetic fields. <i>J. Math. Phys.</i> 64 (2023) 062101. (<i>Non-self-adjoint Hamiltonians: pseudo-Hermiticity without quasi-Hermiticity</i>.) M. F. Hoque, O. Kubů, A. Marchesiello, L. Šnobl. New classes of quadratically integrable systems with velocity dependent potentials: non-subgroup type cases. <i>Eur. Phys. J. Plus</i> 138 (2023) 845 O. Kubů, A. Marchesiello, L. Šnobl. Superintegrability of separable systems with magnetic field: the cylindrical case. <i>J. Phys. A</i> 54 (2021) 425204 S. Bertrand, O. Kubů, L. Šnobl. On superintegrability of 3D axially-symmetric non-subgroup-type systems with magnetic fields. <i>J. Phys. A</i> 54 (2021) 015201 O. Kubů, L. Šnobl. Superintegrability and time-dependent integrals. <i>Archivum Math.</i> 55 (2019) 309–318
SELECTED TALKS	Invited, <i>Geometry and Integrable Systems</i>, La Trobe University, Melbourne (Feb 2026); Invited, <i>Superintegrability, Exact-Solvability, and Representation Theory</i> (Nov 2024); Seminar, Università degli Studi di Milano (Feb 2026); <i>XXXIII International Fall Workshop on Geometry and Physics</i> (Sep 2025). Full list on the website.